

Micah Lee McCollum

micahm@umd.edu

m-lm.github.io

linkedin.com/in/micahmccollum

EDUCATION

University of Maryland, College Park

Aug 2026 - Present

PhD in Computer Science

- Dean's Fellowship

University of Arkansas, Fayetteville

Aug 2023 - May 2026

BS in Computer Science, *summa cum laude*, minor in Mathematics

- Honors Thesis: Towards Multi-Hop Retrieval Using Bipartite Question-Oriented Graphs
- Advisor: Dr. Susan Gauch
- GPA: 3.92/4.00

University of Arkansas, Pulaski Technical College

Aug 2021 - May 2023

AS in Technology & Engineering

- GPA: 4.00/4.00

EXPERIENCE

University of Arkansas, Fayetteville

Apr 2025 - Sep 2025

Research Assistant

Advisor: Dr. Susan Gauch

- Researched knowledge graph-based retrieval-augmented generation (RAG) for grounding LLMs structured context.
- Conducted literature review on knowledge graphs, question answering, and RAG; refactored experimental Python code.

University of Arkansas, Fayetteville

May 2025 - Jul 2025

Teaching Assistant

- Assisted in administering CSCE 20004: Programming Foundations I accelerated online course as the teaching assistant to 30+ students.
- Graded weekly C++ programming projects promptly, provided student feedback, supported course logistics, and held office hours.

PUBLICATIONS

Micah McCollum and Susan Gauch, "Improving Multi-Hop Retrieval for Question Answering via Bipartite Question-Oriented Graphs", *The Eighteenth International Conference on Information, Process, and Knowledge Management*, 2026.

PROJECTS

BiQOG github.com/m-lm/biqog

- Performed experiments for an honors thesis, evaluating methods against baselines to improve multi-hop retrieval in RAG systems.
- Designed and implemented BiQOG, a bipartite question-oriented graph method that augments hypothetical question generation from document passages by explicitly modeling question-passage relationships.

LLM-Powered Cybersecurity Repository

- Co-developed a centralized, searchable repository for cybersecurity research as part of my senior capstone with a team of peers under the sponsorship of Dr. Qinghua Li.
- Served as liaison between the team and sponsor and initiated customer discovery with prospective users to scope use cases and align requirements.
- Designed and built the RAG pipeline, implementing hybrid full-text and semantic search with Reciprocal Rank Fusion, Maximal Marginal Relevance, and reranking.
- Set up the PostgreSQL database and designed its schema, integrating pgvector with native text search.
- Implemented the Flask backend API and helped design the frontend user experience.
- Deployed a Microsoft Azure Linux virtual machine for application testing and deployment.

Key-Value Server github.com/m-lm/taproot

- Built an in-memory key-value store in C++ loosely inspired by Redis with user configuration and minimal dependencies.
- Supports data persistence via append-only logs with basic compaction, LZ4 compression, and binary serialization.
- Implemented a client-server architecture via TCP sockets and automated builds and deployment with shell scripts.
- Improved speed of data compression by 21x, file writes by 1.7x, and log compaction by 1.4x over the course of development.

Minmath github.com/m-lm/minmath

- Developed and deployed a full stack web app to improve mental math skills.
- Implemented user accounts, leaderboards, personal statistics, gameplay settings, and real-time feedback with a clean minimalist interface for a streamlined experience.
- Built with JavaScript/HTML/CSS on the frontend and Django on the backend, utilizing a Docker container for PostgreSQL during development.

AWARDS & HONORS

Senior of Significance, <i>University of Arkansas</i>	2026
Taft, O'Neal, Geels Scholarship, <i>University of Arkansas</i>	2024 - 2026
Published in <i>Milestones Academic Journal</i> vol. 17, <i>Pulaski Technical College</i>	2022
Math Scholar Award, <i>Pulaski Technical College</i>	2022
Outstanding General Chemistry II Student Award, <i>Pulaski Technical College</i>	2022

VOLUNTEER ACTIVITIES

MATHCOUNTS, <i>MATHCOUNTS Foundation</i>	2026
Hour of Code 2025, <i>Code.org</i>	2025